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Automation

“REPLACING HUMANS With ROBOTS Is NOT A FORMULA For Calculating ROI”

Automation is the next big industrial trend that will drive factories and production units. Baishakhi Dutta speaks with Soni Saran Singh, executive director of NMTronics India Pvt Ltd, about the practical capex and opex of such technologies, trade-offs associated with investments and more

Q. How popular do you think automation is in India?

A. Automation will be the key to success in terms of efficiency for almost every industry. In the electronics manufacturing industry, collaborative robots (cobots) are ready to work alongside humans. But these will be high-tech and quite expensive; therefore ROI will always be a challenge.

For the Indian manufacturing industry, low-cost robots will play a major role. Single-arm SCARA (selective compliance articulated robot arm) robots have a great potential right now. We have supplied more than 100 robots to a leading cellphone company in India where these are being used for screw insertion, battery compression and radio frequency testing.

Q. Do you think the Indian industry is ready for unmanned work floors?

A. This concept is specific to Industry 4.0 where machine-to-machine (M2M) communication will happen, and one machine will control the other. Only when this happens, you will have unmanned work floors.

For example, we need light on the shop floor for humans to see things. But in the absence of humans, there is no need for light. This is called the dark factory. There is a long way to go before this happens in India.

For the electronics industry, pick-and-place companies will be on top. A long time ago, Surface Mount Equipment Manufacturers Association (SMEMA) had derived a standard for connecting conveyers and flow of conveyers from the make of one machine to another. Now, data ports are open for all machines, and you can see and collect data but cannot control it. So, control is going to be the key point for effective M2M communication.

Besides, product changeover from A to B is time-consuming and leads to downtime. Every equipment company is looking to bring the downtime to zero, which translates to zero headcount.

Q. How much capex and opex increase is likely for a typical mid-scale automated manufacturing facility?

A. Manufacturing plants are becoming more efficient. When you talk about capex, it is more or less the same as two years ago, but speed and capacity have gone up drastically.

When you look at opex, machines are becoming stable and require little manual intervention. So, while capex remains the same, but with machinery becoming more efficient, cost of opex is going down with investments in machines.

The biggest concern in India is that labour cost is going up. Hence, more automation is required, and the industry needs to accept these changes.

Q. What ROI does NMTronics assure to customers on automation?

A. There is no fixed formula for ROI, for it varies from company to company. ROI depends on the costing of a specific company and what level of automation a customer is looking at.

However, it typically takes two to four years for any type of automated work floor to generate ROI. If you only look at the cost of ROI with replacing humans with robots, then ROI will be long. What you must see are the direct and indirect benefits of bringing robots to the production floor and automation to the production line. That will define ROI. Replacing humans with robots is not a formula for calculating ROI.

The number of defects may go down as, once programmed, robots do not tend to make mistakes. But human

operators can or may leave an organisation. Then, training new operators costs money. There are many indirect things. The robotics industry is highly matured currently.

Q. How is maintenance taken care of in an Industry 4.0 setup?

A. Industry 4.0 setup is completely M2M. There is no such thing as necessary when it comes to maintenance. It is just a matter of how much data is being collected.

Industry 4.0 is basically bringing all data into one place. It will also lead to predictive maintenance based on the data collected.

Q. Is security a concern in an unmanned workforce?

A. Data security is a concern in the world right now. When you are a single site, data can be secured easily. But when you have multiple sites, and data is flowing in the cloud, there are many possible breach points for hackers.

Hackers are looking for data like credit card details, which they can monetise. For SMT machineries, you collect SMT data and focus on SMT side. This data will not be of interest to everyone. But when it comes to the final product, data may be useful. But there are ways to secure it.

Q. How are SMT players promoting the Internet of Things (IoT) ecosystem?

A. SMT players are active and, while proposing Industry 4.0, they say that almost every machine is ready for the smart factory. Everyone is tying up with everybody and gearing up to share data among each other. Data processing is a tool where everyone is a customer. Data collection points, that is, data ports, are now getting open. You can collect my data, but you cannot touch it. That is a key part here. **EFY**